

Coupled differential equations

Key Points

As with all coupled/simultaneous equations we need to eliminate a variable. To do so we often differentiate one equation and substitute the other

Basic Skills

Q1. Differentiate $\frac{dy}{dx} = 3x + 2y$ with respect to x

Q2. Find the general solution to $\frac{d^2y}{dt^2} - \frac{dy}{dt} - 6y = 0$

Q3. For the coupled ODEs $\frac{dy}{dt} = x - 2y$ and $\frac{dx}{dt} = x + 4y$

a) Show that $\frac{d^2y}{dt^2} + \frac{dy}{dt} - 6y = 0$ and hence find general $y(t)$

b) Hence, by substitution find general $x(t)$

c) Apply the initial conditions $y(0) = 0, x(0) = 1$

d) Sketch $(x(t), y(t))$

e) Describe the behaviour of the solution

Competency Skills

Q4. For the following coupled ODEs find both $x(t), y(t)$ and sketch the behaviour

a) $\frac{dy}{dt} = 2x - 3y$ and $\frac{dx}{dt} = -2x + y$
At $t = 0, x = 1, y = 1$

b) $\frac{dy}{dt} = x - 2y - 5$ and $\frac{dx}{dt} = y - 3x + 4$
At $t = 0, x = 0, y = 2$

Q5. A chemical reaction between compounds A and B take place and are modelled by the differential equations $\frac{dA}{dt} = 2B - 9A + 14, \frac{dB}{dt} = 7A - 3B$

a) What does the +14 mean in the rate for A

b) Find the amount of compound A and B when the reaction stops

c) Solve the coupled ODEs and state what the behaviour is for any initial condition

Advanced Skills

Q6. In a rabbit-wolf model the growth of rabbits depends on how many rabbits and how many wolves (with less wolves the better) whereas the growth of wolves only depends on the number of rabbits

a) For a woodland with $R(t), W(t)$ rabbits and wolves respectively we have $20 \frac{dR}{dt} = 2R - W, 100 \frac{dW}{dt} = R$ find the general solutions for the number of rabbits and wolves

b) Given the population in 1990 were recorded as 2000 rabbits and 30 wolves find the number of rabbits and wolves according to the model in 2030

c) Plot on a 2D graph both the rabbit and wolf population as coordinates and describe the behaviour

d) Decide if the model seems fit for purpose

Extension

Q7. Solve the system $\frac{dx}{dt} = 2z, \frac{dy}{dt} = y - z, \frac{dz}{dt} = x$

Hint: Eliminate two variables